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RESEARCH-ARTICLE

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WANROU HU, Shenzhen University, Shenzhen, Guangdong, China

LI ZHANG, Shenzhen University, Shenzhen, Guangdong, China

GAN HUANG, Shenzhen University, Shenzhen, Guangdong, China

LINLING LI, Shenzhen University, Shenzhen, Guangdong, China

ZHIGUO ZHANG, Shenzhen University, Shenzhen, Guangdong, China

ZHEN LIANG, Shenzhen University, Shenzhen, Guangdong, China

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An Efficient EEG Microstate Analysis Method for Emotion Study

Wanrou Hu

School of Biomedical Engineering,
Health Science Center, Shenzhen
University, Shenzhen, 518060,
China; Email:
huwanrou2019@email.szu.edu.cn

Li Zhang

School of Biomedical Engineering,
Health Science Center, Shenzhen
University, Shenzhen, 518060, China;
Email:
lzhang@szu.edu.cn

Gan Huang

School of Biomedical Engineering,
Health Science Center, Shenzhen
University, Shenzhen, 518060, China;
Email:
huanggan1982@gmail.com

Linling Li

School of Biomedical Engineering,
Health Science Center, Shenzhen
University, Shenzhen, 518060, China;
Email:
lilinling@szu.edu.cn

Zhiguo Zhang

School of Biomedical Engineering,
Health Science Center, Shenzhen
University, Shenzhen, 518060, China;
Email:
zgzhang@szu.edu.cn

Zhen Liang*

School of Biomedical Engineering,
Health Science Center, Shenzhen
University, Shenzhen, 518060, China;
Email:
janezliang@szu.edu.cn

ABSTRACT

Emotion plays an essential role in human health and daily life. Estimating emotions in a brain-level dynamic approach helps to understand the underlying neural mechanism, deepen emotion interpretation, and boost the development of affective computing technology for practical application. EEG microstate analysis is a powerful neurophysiological tool for dynamic EEG characterization, covering both temporal and spatial information of brain activities. In this paper, EEG microstate analysis is introduced for the dynamic analysis of video-evoked emotions. A sequential clustering process is proposed for validated and representative microstates detection for emotion-related EEG dynamics characterization, and the underlying neural activation patterns under different emotion states are explored. A study of emotion-related electrophysiological mechanisms is conducted for investigating the emotional perception and processing in the brain responses. The results demonstrate that EEG microstates extracted from the proposed sequential clustering are discriminative for dynamic emotion analysis. Besides, the dynamically evoked emotions can be effectively described by the activation patterns of EEG microstates, where an increased activation of MS2 and MS4 but decrease activation of MS3 are found after emotion induction. Furthermore, distinct emotional-level effects for valence and arousal are observed, where MS4 activities are negatively associated with valence level, and MS3 activities are positively associated with arousal level. In all, our work validates the possibility of applying EEG microstate analysis for emotion-related neural mechanism investigation. It has also proved EEG microstate analysis is a powerful tool for exploring spatial-temporal brain changes through emotion perception.

*Corresponding author: Zhen Liang, janezliang@szu.edu.cn.

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CCS CONCEPTS

• Theory and algorithms for application domains; • Human computer interaction (HCI); • Life and medical sciences;

KEYWORDS

EEG microstate analysis, dynamic emotion analysis, spatial-temporal brain dynamics, video evoking, emotion neural mechanism, emotion perception

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1 INTRODUCTION

Emotion is a dynamic psychological reaction that plays a crucial role in daily task manipulation, such as decision making, interaction, perception, and mental health maintenance [1, 2]. However, due to the nature of emotion complex and variability, there are still many uncertainties about emotion mechanism [3]. Electroencephalography (EEG) is a non-invasive neural imaging technology with high time resolution and fast data transmission and has been highly recommended for emotion studying [4]. Captured the dynamic changes of brain activities in response to emotional stimulation, dynamic emotion-EEG analysis can serve as an efficient and reliable approach for brain-level mechanism exploration and real-time emotion monitoring development [1, 4].

EEG microstate analysis has been recognized as a powerful neurophysiological tool for describing spatial-temporal changes in brain electrophysiological states. EEG microstate analysis provides insights into the spontaneous activation of global neuronal activity by characterizing the synchronized distribution of electric potential over the scalp in a millisecond time resolution. Moreover, simultaneous EEG-fMRI studies reveal a significant electrophysiological association between EEG microstates and fMRI functional brain networks. Changing microstate activities functionally represent the activation of large-scale brain networks [5]. Notably, EEG microstate

analysis overcomes the delayed effect and slow hemodynamics response in fMRI analysis [6]. Hence, EEG microstate analysis can serve as an alternative method for monitoring spontaneous brain activities and reflecting the functional state of large-scale neuronal networks.

Compared to other traditional EEG analysis methods, EEG microstate analysis is more potential for dynamic emotion analysis, which provides a global investigation on the momentary and functional state of brain activity from a joint spatial-temporal perspective. Focusing on the neural activity of emotions in terms of EEG microstates, an investigation of emotion dynamics has been conducted. For example, Gianotti et al. [7] conducted emotion-evoked microstate analysis on ERP data and found valence-associated microstates responded earlier than arousal-associated ones. Their results explored the effectiveness of microstate-based dynamic emotion analysis. On the other hand, Shen et al. [8] explored the feasibility of using microstate features for emotion classification modeling. Based on a set of public microstate templates, EEG microstate features were extracted for emotion classifier modeling. Their results presented that EEG microstate analysis is an alternative method for affective computing. However, due to the complexity and variability of emotions, how to extract representative microstates in a data-driven manner for emotion-related brain activation patterns representation is still an open issue to address.

In this research, a sequential spatial clustering is proposed for representative EEG microstates extraction with an application to emotion dynamics analysis. An optimal set of the most dominant microstates is extracted from two-step clustering on a public EEG-emotion database (DEAP [9], a Database for Emotion Analysis using Physiological Signals) at a within-subject and cross-subject manner, respectively. In such a data-driven process, the extracted microstates perform well in describing emotion-related brain activities with good toleration of individual differences. Sequentially, spontaneous brain activities under evoked emotions can be represented as changing sequences of microstate responses. The dynamic changes of microstate activities are then measured as neural quantification of emotions. Emotion-evoking task effect and emotional level effect on EEG microstate dynamics are explored, and brain activity patterns under various emotional states are characterized. Here, the emotion-evoking task effect measures microstate activity differences between pre-stimulus and post-stimulus stages. Electrophysiological differences led by emotional state changes are considered. Besides, the emotional level effect detects the differences between low-level and high-level emotional states from valence and arousal dimensions, respectively. In summary, EEG microstate analysis is introduced for dynamic emotion analysis. Our work mainly focuses on exploring the pattern differences in microstate activities during emotion triggering. It is expected to provide insights into emotion-related neural dynamics and investigate how the human brain perceives emotions.

2 METHODS

EEG Data

DEAP database is a publicly well-known physiological dataset for emotion analysis [9]. Forty carefully selected music videos with a fixed length of 1 min were randomly presented to 32 subjects for

emotion triggering. Simultaneously, EEG data were collected by the 32-electrode Biosemi ActiveTwo system at a sampling rate of 512 Hz. After each emotion-evoking task, self-assessment in terms of a 9-point SAM scale was conducted for subjective ratings about the evoked emotion states (as shown in Figure 1). In this study, self-assessment ratings on valence (reflection on pleasantness) and arousal (reflection on excitation) dimensions were adopted for emotional state quantification. Therefore, 1280 trials (32 subjects $\times$ 40 trials) of emotion-related EEG data were adopted to study the dynamic changes of brain activity triggered by continuous emotional videos.

2.1 EEG Preprocessing

Data preprocessing is first performed to remove the unrelated artifacts (e.g., ocular movement, cardiac activity, muscular activity, respiration) and enhance data quality for reliable results. A standard EEG preprocessing procedure was carefully operated on every trial of EEG recordings from the DEAP database. **Step 1:** a bandpass filter in the frequency range of 1 - 45 Hz was applied for filtering out unwanted artifacts, and a notch filter at 50 Hz was designed for power line artifact removal. **Step 2:** noisy electrodes were manually screened out and interpolated by the neighboring 3 electrodes. **Step 3:** a common average re-reference was implemented for zero-mean random noise removal, where EEG data were then adjusted to a reference-free potential distribution. **Step 4:** independent component analysis (ICA) was conducted for ocular artifact component removals. Meanwhile, noisy components caused by body movement or muscular activity were also manually identified and rejected. The clean EEG signals were reconstructed from the remaining signal components. **Step 5:** segmentation was finally applied to divide preprocessed EEG data into three parts: **3s pre-stimulus data**, **60s video-stimulated data**, and **3s post-stimulus data** for the following analysis.

2.2 EEG Microstate Analysis

A standard EEG microstate analysis mainly includes **microstate template detection**, **microstate-based EEG re-representation**, and **microstate feature extraction** [10-12]. The detailed procedure is illustrated below in detail (as the flowchart shown in Figure 2).

Microstate template detection. Candidate EEG topographies with high signal-to-noise are first extracted according to the calculated global field power (GFP) values. GFP measures the electric potential distribution over the scalp at a specified sample point, and the fluctuation of the GFP curve represents the corresponding changes in brain activity [13], given as:

$$GFP = \sqrt{\frac{1}{N} \sum_{i=1}^N (V_i(t) - \overline{V(t)})^2} \quad (1)$$

where $V_i(t)$ stands for the voltage of electrode i at the specified sampling time t . $\overline{V(t)}$ is the average instantaneous field power across global electrodes, and N is the electrode numbers. Previous researches discovered that EEG topographies at the local maxima of the GFP curve had high signal-to-noise and low global dissimilarity [14]. These topographies served as the candidate topographies for further microstate searching. **A sequential clustering**

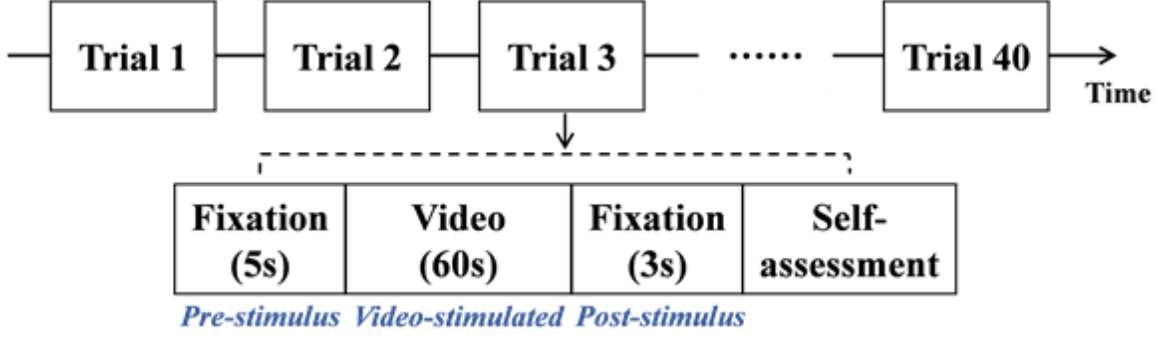


Figure 1: Experimental design of emotion-evoking in the DEAP database.

method is proposed for subject-generalized microstate template detection. Afterward, the original EEG data can be re-represented into changing microstate sequences to capture the dynamic changes of spontaneous brain activities. The detailed procedures of the proposed sequential clustering analysis are presented in 1 (refers to the schematic flowchart in Figure 3).

Algorithm 1 Sequential Clustering Analysis Method

Input: segmented EEG data $E(t, s, c)$

Loop s_j, c_z in Subjects, Conditions

Loop t_i in Trials

Extract candidate topographies from $E(t_i, s_j, c_z)$, and store in $G(s_j, c_z)$

End

detect optimal microstate templates from $G(s_j, c_z)$, and store in $T(s_j, c_z)$

End

detect optimal microstate templates from $T(s_j, c_z)$, and store in *Template*

Output: *Template*

The microstate detection is first conducted on a **within-subject level** for extracting subject-specified EEG microstates at pre-stimulus, video-stimulated, and post-stimulus stages, respectively. For each subject, candidate topographies $G(s_j, c_z)$ from 40 trials of EEG recordings $E(t_i, s_j, c_z)$ under the same experimental conditions are firstly extracted and then put into a modified k-means clustering [10, 13] for optimal topography extraction. For spontaneous EEG data, similar candidate topographies (regardless of polarity inversion) share an equal potential configuration that corresponds to the same functional brain state [15]. With the pre-defined microstate classes k ranging from 2 to 8 and iteration times M of 1000, the initialized centroids for microstate topography clustering are randomly assigned and updated for stable cluster centroids searching. In the iteration of spatial clustering, global explained variance (GEV) and cross-validation criterion (CV) [13] are calculated and updated as indexes of clustering fitness, given by

$$\text{GEV} = \frac{\sum_{t=1}^T (\text{Corr}(V(t), V_c(t)) \cdot \text{GFP}(t))^2}{\sum_{t=1}^T \text{GFP}^2(t)} \quad (2)$$

where $V(t)$ is the potential topography at sampling time t , and $V_c(t)$ refers to the EEG microstate assigned to this time point. $\text{Corr}(\cdot)$ represents the calculation of spatial correlation between two given topographies. $\text{GFP}(t)$ denotes as the global field power of EEG data at a time point t . L is the total time points of EEG recordings. And CV value is calculated as

$$\text{CV} = \hat{\sigma}^2 \cdot \left(\frac{N-1}{N-1-k} \right)^2 \quad (3)$$

where $\hat{\sigma}^2$ is the residual variance estimator [16] calculated as: $\hat{\sigma}^2 = \frac{\sum_t^L V^T(t)V(t) - (V_c^T(t)V(t))^2}{L(N-1)}$. Here, k stands for the number of microstate classes. Through a data-driven approach, the optimal set of EEG microstates are detected as cluster centroids with fewer topographical classes and higher explained variance (high GEV and low CV values). The subject-specific microstate topographies $T(s_j, c_z)$ are extracted from 40 trials. Subsequently, the extracted topographies $T(s_j, c_z)$ are then adopted for **cross-subject** clustering. With the same parameter settings in the first-step clustering, the second-step clustering is performed with the modified k-means clustering, and an optimal set of 4 EEG microstates (*Template*) is identified based on the obtained GEV and CV values.

Microstate-based EEG re-representation. EEG topographies with similar electric potential distribution share approximate neuronal activation patterns. Spatial EEG analysis for microstate assignment offers a new quantification approach to study the whole-brain activation patterns. Based on the identified microstate templates (*Template*), we calculate the spatial relevance of topographies at each EEG sample to microstates. Global map dissimilarity (GMD) is a common measurement of topographical configuration difference [17] by calculating the spatial Pearson's correlation coefficient. Here, GMD is used as the criterion for microstate assignment, calculated as

$$\text{GMD} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left\{ \frac{p_i - \bar{p}}{\sqrt{\frac{1}{N} \sum_{i=1}^N (p_i - \bar{p})^2}} - \frac{q_i - \bar{q}}{\sqrt{\frac{1}{N} \sum_{i=1}^N (q_i - \bar{q})^2}} \right\}^2} \quad (4)$$

where p_i and q_i are the voltage of electrode i in the topography map p and the topography map q , $\bar{p}$ and $\bar{q}$ are the average voltages across global electrodes of p and q . Each EEG time point is then assigned to one microstate class with the highest spatial relevance with

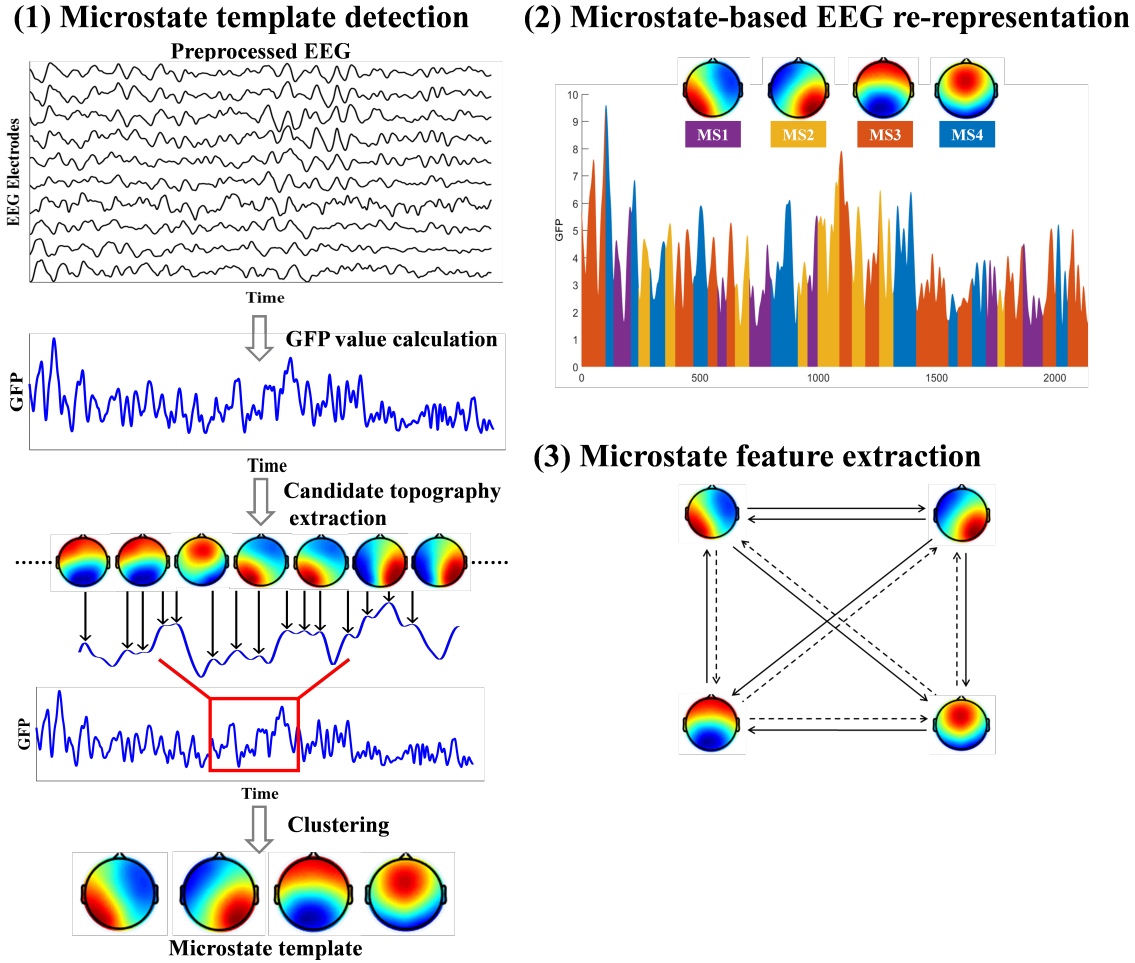


Figure 2: The process of EEG microstate analysis. (1) Microstate template detection. Candidate topographies with quasi-stable topographical configuration are sequentially detected from the local maximal GFP. The GFP value measures the potential distribution of the preprocessed EEG data. Thus, the topographies with higher GFP are presentative for momentary brain state presentation. Based on the extracted topographies, a spatial clustering is adopted for representative microstate templates detection. (2) Microstate-based EEG re-representation. The potential similarity is measured between whole-brain EEG activities and the identified EEG microstate templates. Each EEG time point is then assigned to one microstate with the highest spatial relevance. Therefore, the original EEG time series are re-represented as EEG microstate sequences covering the spatial changes of EEG electrical activities over time. (3) Microstate feature extraction. Statistical features of EEG microstate sequences are extracted for dynamic brain activities quantification.

microstate templates (the highest GMD). After microstate assignment, the original EEG data are represented as changing microstate sequences for emotion-related neurophysiology investigation.

(3) Microstate feature extraction. A bunch of statistical features is calculated from microstate sequences for emotion-related brain dynamics analysis. In this study, the duration, occurrence, coverage, and transition probability of microstate activities are extracted for further analysis. Here, duration measures the average time that a specific microstate remains dominant, reflecting the stability of potential configuration. Occurrence calculates the average number of appearances per second that a specific microstate

topography stays dominant. It provides detection into the preferred patterns of neural coactivation. Coverage measures the ratio of a specific microstate remaining as dominant to the total recording time. Transition probability measures the transition percentage that one dominant microstate transforms into another configured microstate, providing information about brain dynamics in a millisecond time scale [11].

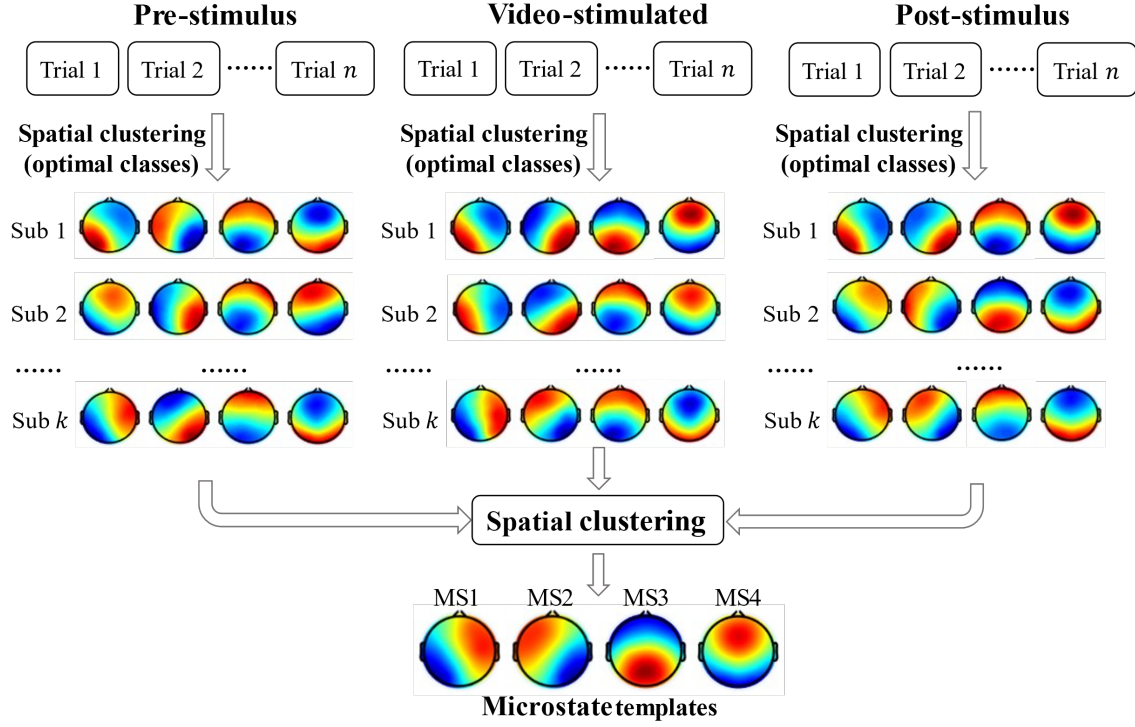


Figure 3: Schematic overview of sequential clustering analysis for microstate template detection. The first-step clustering is conducted at a within-subject level for subject-based representative microstates extraction. Another cross-subject clustering is then adopted for final microstate template extraction. Modified by subject-based information, the identified EEG microstates function for describing the common activation patterns of emotion-related brain responses among 32 subjects.

2.3 Emotion-related analysis on EEG microstates

To estimate the performance of the extracted EEG microstates on characterizing the spatial-temporal changes in EEG activities, a standard evaluation in terms of GEV and CV values measures the fitness of capturing the dynamic changes in emotion-evoked brain dynamics. Subsequently, the activation patterns in EEG microstate sequences are measured to investigate the electrophysiology differences of evoked emotions from the perspective of emotion-evoking task effect and emotional level effect. Inspecting the microstate dynamics provides a new method for dynamic emotion analysis considering changes in brain activities with different evoked emotion states.

(1) Emotion-evoking task effect: measures the microstate activation differences between pre-stimulus and post-stimulus stages. As we all know, task manipulation directly influences brain activity patterns. In line with the experimental design of the DEAP database, pre-stimulus and post-stimulus stages are two resting states before and after emotion-evoking tasks. Therefore, EEG microstate feature differences between pre-stimulus and post-stimulus stages are measured as the index of the emotion-evoking task effect. It examines whether EEG microstates can characterize the brain electrophysiology changes caused by emotion-evoking. The emotion-evoking

task effect on microstate activities is measured by two-step statistical analysis. Such activation differences are first considered at a **within-subject** level to reveal the patterns of subject-specific microstate responses. For each subject, the microstate feature differences before and after emotion-evoking tasks are calculated ($x_k = f_k^{pre} - f_k^{post}$) and adopted for t-statistical analysis, given as $t_k = \frac{\bar{X}_d}{S_d/\sqrt{n}} \cdot \bar{X}_d$ and S_d are the mean and standard deviation of paired-wise microstate feature differences across 40 trials for subject k ($k = 1, \dots, 32$). n denotes the total pairs of microstate features, equals to the number of experimental trials for every subject. Hence, a set of t values $\{t_1, t_2, \dots, t_{32}\}$ is obtained as a measurement of task effect on subject-based microstate activities. To further examine whether the changing patterns of emotion-evoked microstates are consistent across 32 subjects, a two-tailed one-sample t-test is then conducted on the extracted t values at a **cross-subject** approach. The corresponding p-values are corrected for multiple comparisons, applying the false discovery rate (FDR) adjustment with a significant level of 5%.

(2) Emotional level effect: measures the microstate response differences between low- and high-level emotional states, from valence and arousal dimensions, respectively. Based on the 9-point self-assessment ratings on valence and arousal dimensions, a binary

emotional grouping is conducted by adopting a self-adaptive threshold reassignment as described in [18]. Therefore, video-stimulated EEG recordings are divided into two-level emotional groups: low valence vs. high valence, low arousal vs. high arousal. Subsequently, the microstate patterns over different emotional dimensions are investigated. And the emotional level effect on microstate activation differences is measured to interpret the perception changes of emotion processing. A two-tailed Wilcoxon Rank Sum test (non-parametric independent t-test) is implemented to quantify the inter-group differences at every emotional dimension for each microstate feature.

3 RESULT AND DISCUSSION

3.1 Emotion-related microstate template detection

Through the proposed sequential clustering analysis, an optimal set of emotion-related templates with four canonical microstates is identified for the DEAP database (term as emotion-related microstate templates). Notably, the extracted microstate templates have the quasi-stable configuration of left posterior-right anterior (termed as MS1 below), left anterior-right posterior (termed as MS2 below), anterior-posterior (termed as MS3 below), and frontal-central (termed as MS4 below) orientation. Moreover, the identified EEG microstates yield a corresponding GEV of 82.23% and a corresponding CV value of 63.33% after normalizing into the range of [0,1] (Figure 4 (a)). These two indexes measure the global topographical representation over EEG microstate time series. The templates with higher GEV and lower CV values are generally extracted as optimal microstates with a fine fitness on capturing the brain's neurological states.

Meanwhile, another two cases of clustering analysis for emotion-related microstates searching are also discussed for comparison. **Case 1** (as shown in Figure 4 (b)): The first-step clustering is conducting at a cross-subject level. All EEG data (from three experimental conditions across 32 subjects) collected from the same video stimulation are put together for trial-based microstate template searching. Subsequently, the extracted topographies are then applied for cross-trial clustering. A set of 4-class EEG microstates (termed as Tem1 to Tem4) with the GEV of 79.78% and CV of 72.3% is obtained. However, a special-configured microstate (Tem4) with anterior-dominant orientation is quite different from the canonical microstates. The corresponding electrophysiology and its relevance with functional brain networks are unclear. **Case 2** (as shown in Figure 4 (c)): Different from Case 1, trial-based microstate templates are separately extracted at pre-stimulus, video-stimulated, and post-stimulus stages. Next, extracted topographies are then put together for the second-step clustering. In this case, the highest GEV of 87.40% (with a corresponding CV of 50.9%) is observed from the identified 4-class EEG microstates (termed as Tem1 to Tem4). However, there exists lower spatial dissimilarity that the identified microstates share similar electrophysiology activation patterns. In other words, the good performance of these templates may be caused by repeated calculation with huge information bias. In comparison with these two microstate clustering analysis methods, the proposed sequential clustering analysis method seems to be more suitable for emotion-related representative microstate templates.

Following, to further verify the feasibility of using EEG microstate analysis in emotion study, we explore the emotion-evoking task effect and emotion-level effect on microstate dynamics based on the identified microstate templates by the proposed method.

3.2 Emotion-evoking task effect

Microstate activation differences between pre-stimulus and post-stimulus stages are reported in Figure 5, for the emotion-evoking task effect study on EEG activities. Through a two-step statistical analysis, we observe that emotional states can be described from the changing patterns of microstate activities. Statistical results present a consistent pattern of EEG microstate responses before and after emotional task stimulation across 32 subjects. A positive task effect is found on MS2 and MS4 that emotion-evoking task manipulation leads to a significant increase in the coverage of MS2 ($p = 0.0463$, FDR) and MS4 ($p = 0.0453$). But a negative task effect is found on MS3 that decreased coverage ($p = 0.0050$), duration ($p = 0.0286$), and occurrence ($p = 0.0454$) of MS3 are observed at post-stimulus stages. Also, a decreased transition probability from MS4 to MS3 is found after emotional tasks. While the transition probability from MS3 to MS2 and from MS4 to MS2 significantly increases. However, no significant emotion-evoking task effect on MS1 activity is found across 32 subjects. In all, we examine that the extracted EEG microstates are representative for detecting emotion-evoked brain activities. The emotion-evoking task effect on microstate activities is mainly found in MS2, MS3, and MS4.

3.3 Emotional level effect

To further investigate neurological activation patterns of emotions, emotional level effect on dynamic microstate activities at video-stimulated stages is analyzed. For the **valence** dimension, a negative association is found between the MS4 activation pattern and valence level (the results are shown in Figure 6). A lower MS4 occurrence is observed in the high valence group ($p = 0.0379$). A greater transition probability from MS1 to MS2 ($p = 0.0261$) is also found in high valence groups. But the transition from MS1 to MS4 significantly decrease at the high-level valence states comparing to low valence ($p = 0.0228$). For the **arousal** dimension, a positive relationship is found between the MS3 activation pattern and arousal state (refers to Figure 7). Comparing the microstate differences between low and high arousal groups, we observe a higher coverage ($p = 0.0147$) and occurrence ($p = 0.0203$) of MS3 at the high arousal state. No such emotional-level effect is found in MS1, MS2, and MS4. The results also reveal a significant arousal-level effect on microstate transition that a higher transition probability from MS2 to MS3 is presented in high arousal groups.

Generally, the results demonstrate that EEG microstate analysis is a powerful tool for characterizing the brain activity changes under different emotional states. Notably, a distinct activation pattern of EEG microstates is found in different emotional dimensions. Valence-based emotional states mainly relate to the changing response of MS4, while arousal is associated with MS3. Moreover, a positive relationship is found between MS3 activity and emotional level of arousal, but a negative association is observed between MS4 activity and emotional level of valence. Our findings are in line with the results of Milz et al. [19] that MS4 mainly responded in

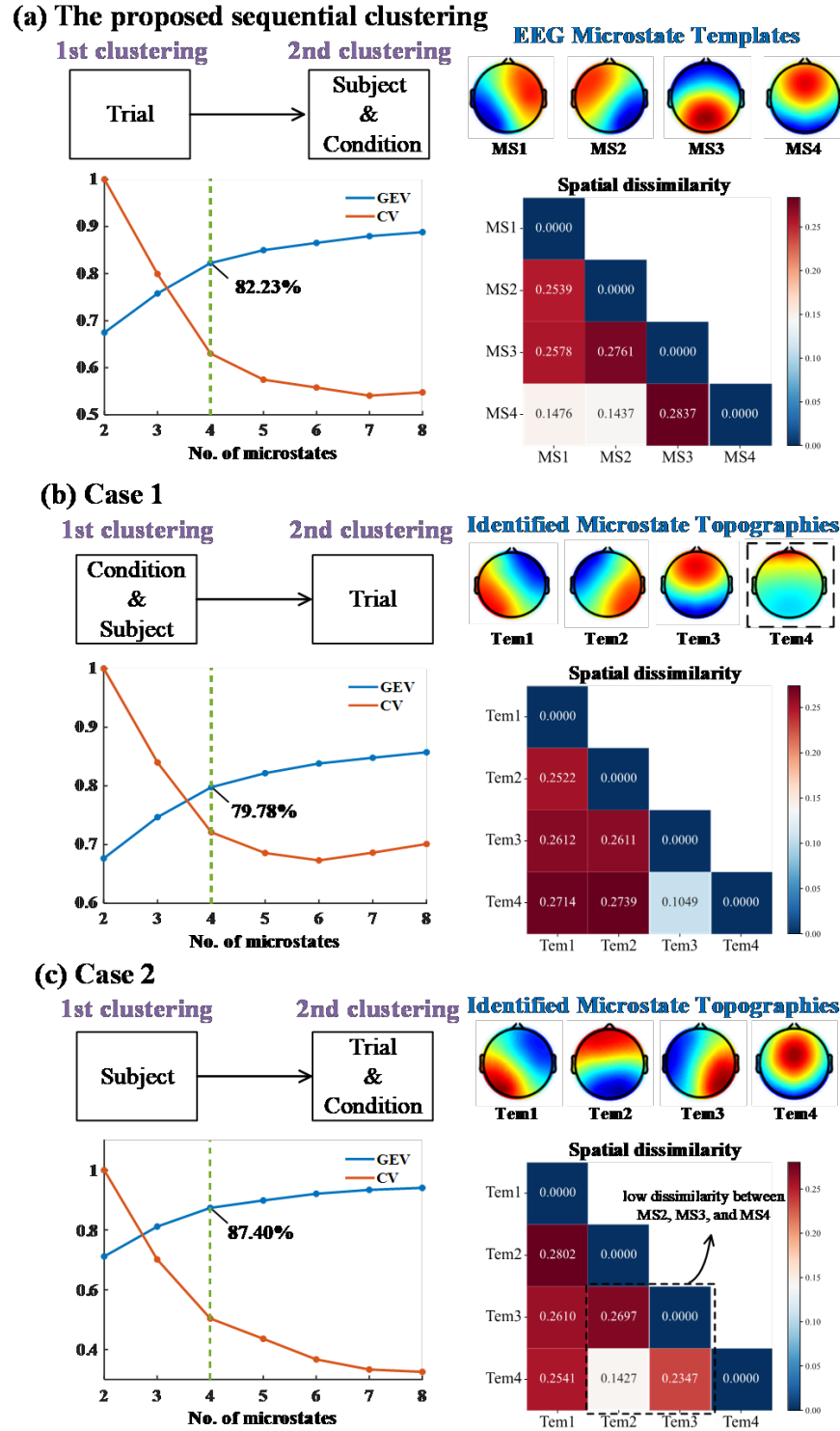


Figure 4: A comparison of EEG microstates identification by the proposed sequential clustering analysis and other two cases of clustering. (a) The explain performance of the processed clustering. (2) The identified EEG microstates in Case 1 correspond to a lower GEV value with low performance for EEG data re-representation. (3) The identified EEG microstate topographies at Case 2 are with a lower spatial dissimilarity. The topographical configurations with high correlation share similar electrophysiological activities, which might not distinctive for describing the specific changing patterns of functional brain networks during emotion triggering.

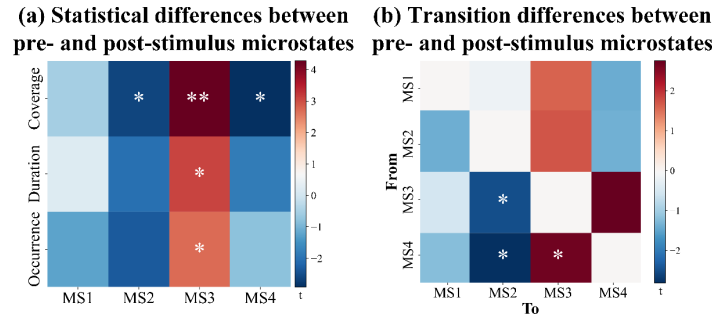


Figure 5: Emotion-evoking task effect on EEG microstate activation patterns. Generally, Statistical differences between pre-stimulus and post-stimulus microstate activities are measured. The t values from the two-tailed statistical analysis are showed that the positive t value (red) indicates higher microstate activities at the pre-stimulus stage than the post-stimulus, and the negative t value (blue) indicates higher microstate activities at the post-stimulus stage. (* $p < 0.05$, ** $p < 0.01$)

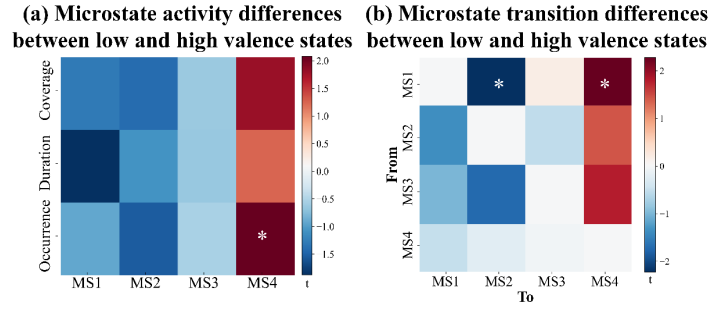


Figure 6: Emotional level effect on EEG microstates at the valence dimension. The microstate response differences between low and high valence states are analyzed for emotion-related neural mechanism investigation. The positive t value (red) from the two-tailed statistical analysis indicates a higher microstate response at the low valence state. And the negative t value (blue) is for a higher response at the high valence state. (* $p < 0.05$)

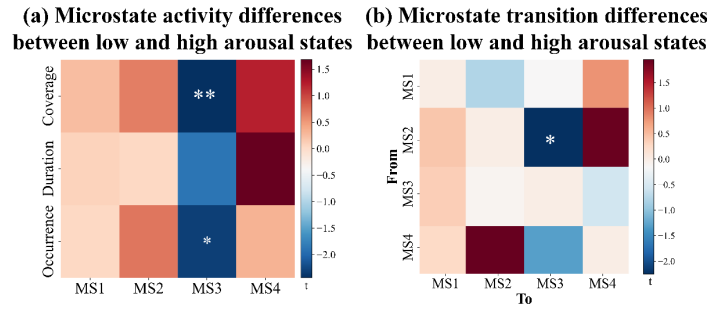


Figure 7: Emotional level effect on EEG microstates at the arousal dimension. The positive t value (red) reveals a higher microstate response at the low arousal state, but the negative t value (blue) declares a higher response at the high arousal state. (* $p < 0.05$, ** $p < 0.01$)

goal-directed perception processing. They also observed decreased coverage, duration, and occurrence of MS4 in goal-directed perception tasks comparing to the resting state. Besides, based on the functional association of MS3 and MS4 to the default mode network and dorsal attention network, our results also suggest a distinct neurological mechanism of emotional perception for valence and arousal

dimensions. In general, the emotional level differences can be represented by the changing patterns of EEG microstate sequences. This study examines the possibility of applying EEG microstate analysis for emotion-related neural mechanism investigation.

The findings suggest that brain activity detected by EEG microstates is dynamically changing in response to continuous stimulations of multimedia videos for emotion triggering. Previous

researches generally revealed that there existed a closed association between four canonical EEG microstates (from MS1 to MS4) and significant functional brain networks (the auditory network, visual network, default mode network, and dorsal attention network) [21, 22]. Therefore, comparing to the traditional EEG analysis methods, EEG microstate analysis can perform well in dynamic emotion analysis for combined detection on spatial-temporal EEG changes and monitor the activation patterns of neurological brain states. For example, Milz et al. [19] found that different task modalities led to varied effects on EEG microstate properties. They separately conducted task-free resting, object-visualization, spatial-visualization, and verbalization tasks for modality-specific thinking induction. Their microstate-based results showed that the parameter alternation of MS1 and MS2 were separately associated with visual and verbal modalities. And the property of MS4 decreased in goal-directed perception tasks comparing to the resting state. Similarly, Seitzman et al. [22] found that cognitive tasks mainly led to a significant increase in MS4 properties (duration, coverage, and occurrence), whereas a significant decrease in MS3; while visual stimulated task led to a significant increase in the occurrence and coverage of MS2 and the occurrence of MS3 during eye-opened. But the duration of MS1 and MS4 during the eye-open period was shorter than that in the eye-closed situation. Those findings demonstrated that the changes in microstate properties benefit quantifying the inherent brain states in a spatial-temporal perspective responding to both internal and external stimuli.

4 CONCLUSION

Triggered by music videos with rich audiovisual stimulations, emotions are dynamically elicited, leading to spontaneous changes in brain signals. Tracing the dynamic brain responses in emotion triggering helps to understand the emotion perceiving process. In this study, EEG microstate analysis is introduced for emotion-related EEG dynamics analysis, reflecting the changes in brain response. A sequential clustering method is proposed for EEG microstate templates detection. The emotion-evoking task effect and emotional level effect on EEG microstate activities are examined. The results show emotions can be effectively characterized by the activation patterns of EEG microstates. Affected by emotion-evoking tasks, an increase of MS2 and MS4 responses but a decrease of MS3 activities are observed at the post-stimulus stage. Moreover, valence-based and arousal-based emotion level differences also lead to a varied alternation in microstate sequences. A positive association between MS4 activities and valence states is observed. Compared to low valence, high valence is corresponding to a higher MS4 activation. Besides, a negative association is found between MS3 activities and arousal levels. A lower MS3 response is observed in the high arousal groups. Overall, our work investigates the possibility of applying EEG microstate analysis for emotion-related neural mechanism exploration. Microstate-based emotion study offers alternative insights into emotion perceiving, which will greatly deepen our understanding of human emotions.

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REFERENCES

- [1] S. M. Alarcão, and M. J. Fonseca, "Emotions recognition using EEG signals: A survey," *IEEE Transactions on Affective Computing*, vol. 10, no. 3, pp. 374–393, Jul. 2019.
- [2] L. Shu, J. Xie, M. Yang, Z. Li, Z. Li, D. Liao, X. Xu, and X. Yang, "A review of emotion recognition using physiological signals," *Sensors*, vol. 18, no. 7, pp. 2074, Jun. 2018.
- [3] W. Hu, G. Huang, L. Li, L. Zhang, Z. Zhang, and Z. Liang, "Video-triggered EEG-emotion public databases and current methods: A survey," *Brain*, vol. 6, no. 3, pp. 255–287, Feb. 2020.
- [4] Y.-J. Liu, M. Yu, G. Zhao, J. Song, Y. Ge, and Y. Shi, "Real-time movie-induced discrete emotion recognition from EEG signals," *IEEE Transactions on Affective Computing*, vol. 9, no. 4, pp. 550–562, Oct.–Dec. 2018.
- [5] F. Musso, J. Brinkmeyer, A. Mobascher, T. Warbrick, and G. Winterer, "Spontaneous brain activity and EEG microstates. A novel EEG/fMRI analysis approach to explore resting-state networks," *Neuroimage*, vol. 52, no. 4, pp. 1149–1161, Oct. 2010.
- [6] D. Vidaurre, S. M. Smith, and M. W. Woolrich, "Brain network dynamics are hierarchically organized in time," *Proceedings of the National Academy of Sciences*, vol. 114, no. 48, pp. 12827–12832, Oct. 2017.
- [7] L. R. R. Gianotti, P. L. Faber, M. Schuler, R. D. Pascual-Marqui, K. Kochi, and D. Lehmann, "First valence, then arousal: the temporal dynamics of brain electric activity evoked by emotional stimuli," *Brain topography*, vol. 20, no. 3, pp. 143–156, Jan. 2008.
- [8] X. Shen, X. Hu, S. Liu, S. Song, and D. Zhang, "Exploring EEG microstates for affective computing: decoding valence and arousal experiences during video watching," in 2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), Canada, 2020, pp. 841–846.
- [9] S. Koelstra, C. Muhl, M. Soleymani, J.-S. Lee, A. Yazdani, T. Ebrahimi, T. Pun, A. Nijholt, and I. Patras, "Deap: A database for emotion analysis; using physiological signals," *IEEE transactions on affective computing*, vol. 3, no. 1, pp. 18–31, Jan. 2012.
- [10] C. M. Michel, and T. Koenig, "EEG microstates as a tool for studying the temporal dynamics of whole-brain neuronal networks: a review," *Neuroimage*, vol. 180, pp. 577–593, Oct. 2018.
- [11] A. Khanna, A. Pascual-Leone, C. M. Michel, and F. Farzan, "Microstates in resting-state EEG: current status and future directions," *Neurosci Biobehav Rev*, vol. 49, pp. 105–13, Feb. 2015.
- [12] J. Liu, J. Xu, G. Zou, Y. He, Q. Zou, and J. H. Gao, "Reliability and Individual Specificity of EEG Microstate Characteristics," *Brain Topogr*, vol. 33, no. 4, pp. 438–449, Jul. 2020.
- [13] R. D. Pascual-Marqui, C. M. Michel, and D. Lehmann, "Segmentation of brain electrical activity into microstates: model estimation and validation," *IEEE Transactions on Biomedical Engineering*, vol. 42, no. 7, pp. 658–665, Jul. 1995.
- [14] W. Skrandies, "Global field power and topographic similarity," *Brain topography*, vol. 3, no. 1, pp. 137–141, Sep. 1990.
- [15] A. Pedroni, L. R. R. Gianotti, T. Koenig, D. Lehmann, P. Faber, and D. Knoch, "Temporal Characteristics of EEG Microstates Mediate Trial-by-Trial Risk Taking," *Brain Topography*, vol. 30, no. 1, pp. 149–159, Jan. 2017.
- [16] S. P. Muthukrishnan, N. Ahuja, N. Mehta, and R. Sharma, "Functional brain microstate predicts the outcome in a visuospatial working memory task," *Behavioural brain research*, vol. 314, pp. 134–142, Nov. 2016.
- [17] D. Brunet, M. M. Murray, and C. M. Michel, "Spatiotemporal analysis of multichannel EEG: CARTOOL," *Comput Intell Neurosci*, vol. 2011, pp. 813870, Jan. 2011.
- [18] J. Fdez, N. Guttenberg, O. Witkowski, and A. Pasquali, "Cross-Subject EEG-Based Emotion Recognition Through Neural Networks With Stratified Normalization," *Front Neurosci*, vol. 15, pp. 626277, Feb. 2021.
- [19] P. Milz, P. L. Faber, D. Lehmann, T. Koenig, K. Kochi, and R. D. Pascual-Marqui, "The functional significance of EEG microstates—Associations with modalities of thinking," *Neuroimage*, vol. 125, pp. 643–656, Jan. 2016.
- [20] J. Britz, D. Van De Ville, and C. M. Michel, "BOLD correlates of EEG topography reveal rapid resting-state network dynamics," *Neuroimage*, vol. 52, no. 4, pp. 1162–1170, Oct. 2010.
- [21] L. Brechet, D. Brunet, G. Birot, R. Gruetter, C. M. Michel, and J. Jorge, "Capturing the spatiotemporal dynamics of self-generated, task-initiated thoughts with EEG and fMRI," *Neuroimage*, vol. 194, pp. 82–92, Jul. 2019.
- [22] B. A. Seitzman, M. Abell, S. C. Bartley, M. A. Erickson, A. R. Bolbeck, and W. P. Hetrick, "Cognitive manipulation of brain electric microstates," *NeuroImage*, vol. 146, pp. 533–543, Feb. 2017.